

HOVDEN-LUNDANE, Norway

WMO: 014410

Lat: 59.5767N

Lon: 7.3897E

Elev: 836

StdP: 91.68

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-26.3	-23.8	-29.0	0.3	-26.3	-26.3	0.4	-23.6	10.3	-2.0	9.4	-2.5	1.7	40	0.741

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	21.4	13.2	19.6	12.3	17.5	11.7	14.3	19.3	13.4	17.7	12.3	16.3	3.2	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	9.8	15.8	11.2	9.2	15.1	10.3	8.6	14.0	42.7	19.2	40.0	17.6	37.3	16.5	17.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.4	6.4	DB	-29.4	23.7	2.5	2.2	-31.2	25.3	-32.7	26.6	-34.1	27.8	-35.9	29.5	(4)
(5)			WB	-29.6	15.4	2.5	1.3	-31.4	16.3	-32.8	17.1	-34.2	17.8	-36.0	18.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.1	-7.4	-7.6	-4.7	-0.2	4.2	8.0	10.7	9.8	6.9	2.1	-2.6	-6.1	(6)
(7)		DBStd	7.99	6.50	6.50	5.47	3.92	3.42	2.66	2.82	2.52	2.92	3.74	5.20	6.91	(7)
(8)		HDD10.0	3331	540	492	456	306	184	71	25	35	101	246	377	499	(8)
(9)		HDD18.3	6275	798	726	714	556	439	310	237	263	344	504	627	757	(9)
(10)		CDD10.0	97	0	0	0	0	3	11	47	30	6	0	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	8	0	0	0	0	0	0	8	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	2.8	3.2	3.0	3.0	2.6	2.6	2.7	2.5	2.5	2.7	2.9	2.9	(14)	
(15)	Precipitation	PrecAvg	1099	124	97	83	53	49	63	77	93	88	108	108	124	(15)
(16)		PrecMax	1594	337	245	336	88	124	148	143	184	171	214	224	314	(16)
(17)		PrecMin	615	8	10	0	13	2	9	16	13	35	15	29	4	(17)
(18)		PrecStd	213	76	67	64	19	26	35	26	37	36	57	57	74	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	5.0	7.6	11.8	20.4	22.0	24.5	21.5	18.3	12.5	8.0	5.4	(19)
(20)			MCWB	2.4	3.0	3.1	6.3	11.2	11.3	14.6	13.6	11.7	7.6	6.5	4.1	(20)
(21)		2%	DB	2.7	3.1	4.7	9.2	16.9	20.0	22.3	19.2	15.7	10.4	6.6	4.2	(21)
(22)			MCWB	1.6	1.9	2.1	4.7	10.2	11.2	14.2	12.8	10.5	7.4	5.2	3.1	(22)
(23)		5%	DB	1.8	2.1	3.4	7.1	13.8	17.6	20.5	17.3	13.8	9.0	5.0	3.1	(23)
(24)			MCWB	0.7	0.8	1.5	3.6	8.2	10.3	13.4	12.2	10.2	6.8	3.8	2.0	(24)
(25)		10%	DB	0.7	0.8	2.3	5.4	11.2	15.3	18.3	15.6	12.2	7.6	3.7	1.9	(25)
(26)			MCWB	-0.2	-0.3	0.7	2.6	6.8	9.2	12.5	11.3	9.4	5.7	2.5	0.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	3.3	3.9	6.8	12.3	13.2	15.9	14.7	13.4	9.3	6.6	4.2	(27)
(28)			MCDB	3.6	4.6	5.9	10.2	18.2	18.0	21.7	19.4	16.8	10.9	7.8	5.1	(28)
(29)		2%	WB	1.8	2.1	2.6	5.2	10.4	12.2	14.9	13.8	11.6	8.0	5.2	3.1	(29)
(30)			MCDB	2.6	3.1	4.1	8.3	15.8	18.0	20.8	17.4	14.0	9.8	6.5	4.1	(30)
(31)		5%	WB	0.8	0.9	1.8	4.0	8.7	11.1	14.0	12.9	10.6	7.0	3.9	2.0	(31)
(32)			MCDB	1.7	1.9	3.3	6.8	12.9	15.8	19.0	16.4	13.1	8.7	5.0	2.9	(32)
(33)		10%	WB	-0.2	-0.3	0.6	2.9	7.3	10.0	13.0	11.8	9.6	5.8	2.6	0.9	(33)
(34)			MCDB	0.7	0.7	2.1	5.1	10.9	14.2	17.5	15.1	12.0	7.4	3.7	1.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.7	8.8	10.1	9.3	9.4	11.2	10.2	9.6	8.6	7.6	6.8	7.8	(35)
(36)			MCDBR	6.8	7.2	9.3	11.5	14.7	17.5	16.7	13.4	12.5	10.1	6.5	5.6	(36)
(37)			MCWBR	6.3	6.2	6.9	7.5	8.5	9.5	8.9	7.8	8.0	7.1	5.7	5.1	(37)
(38)		5% WB	MCDBR	6.8	6.9	8.2	9.9	13.3	13.5	14.0	11.5	9.9	7.6	6.1	5.9	(38)
(39)			MCWBR	6.5	6.1	6.2	6.5	8.1	8.1	7.8	7.1	6.9	6.0	5.5	5.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.219	0.255	0.285	0.321	0.325	0.320	0.335	0.329	0.303	0.285	0.235	0.195	(40)	
(41)		taud	2.168	2.262	2.235	2.235	2.371	2.487	2.458	2.467	2.539	2.489	2.247	2.066	(41)	
(42)		Ebn at Noon	609	775	853	880	904	915	889	869	842	743	581	486	(42)	
(43)		Edh at Noon	41	67	94	114	107	98	98	90	71	54	38	30	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.33	0.95	2.24	4.06	4.68	5.01	4.52	3.54	2.27	1.11	0.41	0.21	(44)	
(45)		RadStd	0.08	0.23	0.48	0.78	0.83	0.62	0.61	0.31	0.23	0.19	0.08	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page